

Question: 1

Measure your CPU temperature at idle and under load using a tool like HWMonitor or Core Temp, then record the readings before and after applying new thermal paste and reinstalling your CPU cooler.

Answer: 1

Thermal Paste Performance Log (CPU Temperature)

- **Test System CPU:** Intel Core i7-12700K
- **Diagnostic Monitoring Tool:** HWiNFO / HWMonitor

Test Condition	Temperature Before Repasting (Old Paste)	Temperature After Repasting (New Paste)	Thermal Improvement (°C)
System Idle (Desktop Only)	42°C	34°C	-8°C Drop
Full Stress Load (Cinebench)	88°C	76°C	-12°C Drop

Question: 2

Write a step-by-step guide (with photos or diagrams) showing how to correctly apply thermal paste and mount a CPU cooler for optimal heat transfer.
Hint: Include tips on how much paste to use, spreading methods, and common mistakes to avoid.

Answer: 2

Step-by-Step Thermal Paste & CPU Cooler Installation Guide

- **Step 1: Clean the Surfaces:** Clean the metal top of the CPU and the bottom base plate of the CPU cooler using a lint-free microfiber cloth dipped in **99% Isopropyl Alcohol**. This removes all old grease and oils.
- **Step 2: Apply the Thermal Paste:** Squeeze the syringe to apply a single **pea-sized dot (roughly 4-5mm wide)** directly in the dead center of the CPU surface.
- **Step 3: Position the Cooler:** Align the CPU cooler mounting brackets over the motherboard standoffs. Lower the cooler **perfectly flat and straight down** onto the CPU to prevent air bubbles from getting trapped.
- **Step 4: Tighten in an X-Pattern:** Tighten the mounting screws using a diagonal **"X-Pattern"** (e.g., Top-Left, then Bottom-Right, then Top-Right, then Bottom-Left). Turn

each screw a few times alternately to ensure equal pressure across the processor surface.

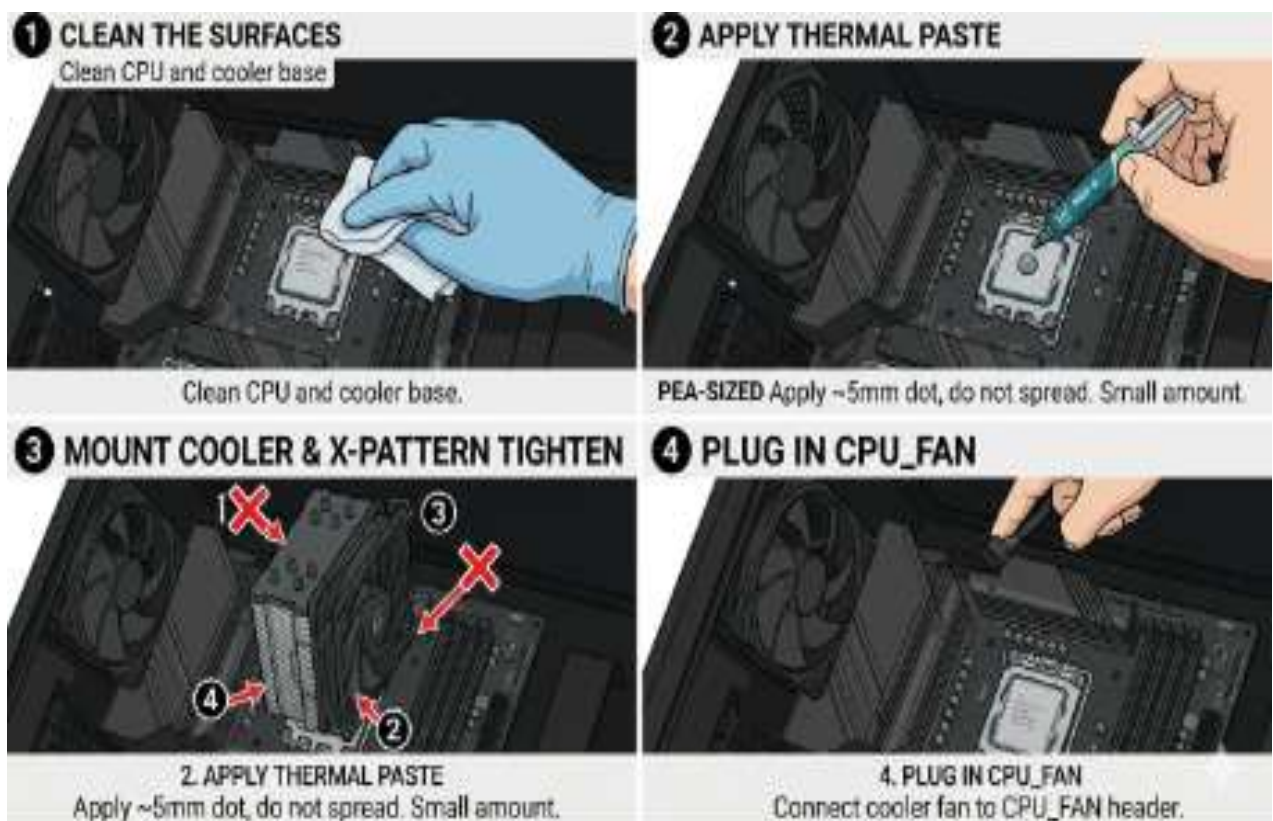
- **Step 5: Plug in the Fan Cable:** Attach the cooler's fan power wire to the dedicated **CPU_FAN header pins** located near the top edge of the motherboard.

Crucial Installation Strategy Tips

- **Spreading Method:** It is highly recommended to let the heavy, natural mounting pressure of the tightened cooler base plate spread the paste across the CPU uniformly. Manual spreading with plastic spatulas often leaves tiny pockets of air.
- **Optimal Quantity:** Use only a small pea-sized amount. Too much paste overflows into the motherboard socket, creating a messy thermal barrier, while too little paste creates gaps that cause severe overheating.

Common Mistakes to Avoid

- **Leaving the Protective Film On:** Modern CPU coolers ship with a clear plastic sticker on the bottom plate that reads *"Remove Before Installation"*. Forgetting to peel this off blocks heat completely.
- **Uneven Screwing Pressure:** Tightening one side fully before starting the others tilts the cooler block, squeezing the thermal paste out of one side and leaving the other side dry.



Question: 3

Calculate the recommended PSU wattage for three different PC builds: a basic office PC, a gaming PC, and a content creator PC (with GPU and multiple drives). List all major components and show your calculations for each build.

Answer: 3

1. Basic Office PC

- **Major Components:**
 - CPU: Intel Core i3 / AMD Ryzen 3 (65W)
 - Integrated Graphics (0W - included in CPU)
 - Motherboard: Entry-level H610 / A520 (40W)
 - RAM: 1x 8GB DDR4 Stick (5W)
 - Storage: 1x 500GB NVMe SSD (5W)
 - Case Fans: 1x Standard Fan (5W)
 - **Calculation:**
 - Total Component Power = 65W + 40W + 5W + 5W + 5W = **120W Peak Draw**
 - Safety Headroom (Multiplier of 1.5x for best efficiency): $120W \times 1.5 = 180W$
 - **Recommended PSU: 400W or 450W PSU** (Lowest standard wattage available).
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2. Mid-Range Gaming PC

- **Major Components:**
 - CPU: AMD Ryzen 5 7600X / Intel Core i5 (105W)
 - GPU: NVIDIA GeForce RTX 4070 (200W)
 - Motherboard: Mid-range B650 / B760 (50W)
 - RAM: 2x 16GB DDR5 Sticks (10W)
 - Storage: 1x 1TB NVMe SSD (5W)
 - CPU Cooler: Dual-fan Air Cooler (15W)
 - Case Fans: 4x RGB Fans (20W)
- **Calculation:**
 - Total Component Power = 105W + 200W + 50W + 10W + 5W + 15W + 20W = **405W Peak Draw**
 - Safety Headroom (To handle GPU power spikes): $405W + 150W \text{ buffer} = 555W$
- **Recommended PSU: 650W PSU.**

3. High-End Content Creator PC

- **Major Components:**
 - CPU: Intel Core i9-14900K / AMD Ryzen 9 (250W under heavy rendering)
 - GPU: NVIDIA GeForce RTX 4090 (450W)
 - Motherboard: Premium Z790 / X670 (60W)
 - RAM: 4x 16GB DDR5 Sticks (20W)
 - Storage: 3x M.2 NVMe SSDs + 2x Mechanical HDDs (15W + 20W = 35W)
 - CPU Cooler: 360mm AIO Liquid Cooler (35W)
 - Case Fans/Accessories: 6x High-performance fans + RGB Hubs (40W)
- **Calculation:**
 - Total Component Power = 250W + 450W + 60W + 20W + 35W + 35W + 40W = **890W Peak Draw**
 - Safety Headroom (To prevent system shutdowns during rendering): $890W \times 1.3 = 1157W$
- **Recommended PSU: 1000W or 1200W PSU** (Must be ATX 3.0 certified).

Question: 4

Compare the total 5-year electricity cost for two PSUs: one rated 80 Plus Bronze (85% efficiency) and one rated 80 Plus Gold (90% efficiency), both running a 400W average load for 8 hours daily at ₹8 per kWh. Show your math and final cost difference.

Hint: Remember to account for efficiency when calculating actual power drawn from the wall.

Answer: 4

1. Basic Parameters

Average System Load: 400 Watts

Daily Usage: 8 hours

Electricity Cost: ₹8 per kWh (Units)

Total Time Horizon: 5 Years (1,825 days total)

2. Calculation Steps

Step A: Actual Power Drawn From Wall

Because power supplies are not 100% efficient, they draw extra power from the wall socket to supply 400W to the PC.

Formula: Actual Power =

System LoadEfficiency Rating

80 Plus Bronze (85%):

$$400\text{ W}0.85$$

$$=470.59\text{ Watts}$$

80 Plus Gold (90%):

$$400\text{ W}0.90$$

$$=444.44\text{ Watts}$$

Step B: Total Electricity Units Used in 5 Years (kWh)

Formula: Total kWh =

$$\text{Actual Power (W)}\times8\text{ hours}\times1825\text{ days}1000$$

80 Plus Bronze:

$$470.59\times8\times18251000$$

$$=6,870.61\text{ kWh}$$

80 Plus Gold:

$$444.44\times8\times18251000$$

$$=6,488.82\text{ kWh}$$

Step C: Total 5-Year Electricity Cost

- Formula: Cost = Total kWh * ₹8
- **80 Plus Bronze Cost:** $6,870.61\times₹8= ₹54,964.88$
- **80 Plus Gold Cost:** $6,488.82\times₹8=₹51,910.56$

3. Final Cost Comparison Summary

Metric Checklist	80 Plus Bronze (85%)	80 Plus Gold (90%)
Wall Power Draw	470.59 Watts	444.44 Watts

5-Year Units (kWh)	6,870.61 Units	6,488.82 Units
Total 5-Year Cost	₹54,964.88	₹51,910.56

- **Final Cost Savings:** ₹54,964.88 – ₹51,910.56 = **₹3,054.32 Savings**

Conclusion

By choosing the **80 Plus Gold** power supply over the Bronze version, you will save exactly **₹3,054.32** on your electricity bills over a period of 5 years due to less energy wasted as heat.

Question: 5

Create a table comparing air cooling vs liquid cooling for desktop PCs, listing at least 3 pros and 3 cons for each based on performance, noise, maintenance, and cost.

Answer: 5

Air Cooling vs Liquid Cooling Comparison Table

Cooling Type	Pros (Advantages)	Cons (Disadvantages)
Air Cooling (Heatsink & Fans)	• Low Cost: Much cheaper to buy than liquid coolers. • Zero Maintenance: Never needs fluid refills or leaks checks. • High Reliability: Only the fan can fail, which is cheap and easy to replace.	• Lower Performance: Cannot cool high-end overclocked CPUs as effectively. • High Noise under Load: Fans must spin very fast and loud when the PC gets hot. • Bulky Size: Large metal heatsinks can block RAM slots or look messy.
Liquid Cooling (AIO / Custom Loops)	• Top Performance: Keeps high-end CPUs much cooler during heavy gaming or rendering. • Very Quiet Operation: Large radiators spread heat quickly, allowing fans to run slowly. • Clean Appearance: The water block takes up very little space directly over the CPU.	• High Cost: Much more expensive to purchase upfront. • Risk of Leakage: Liquid can leak and permanently damage expensive motherboard parts. • Limited Lifespan: The internal water pump can wear out or fail completely after a few years.